

UMC 0.11 um Logic/Mixed_Mode AI Advanced Enhancement RVT 5 V MOSFET Monte Carlo SPICE Model Document.

**Version 0.2 Phase 1
2012/07/09**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/07/09	Wei Lun Hung	This Monte Carlo model includes both process and mis-matching characteristics extracted from model cards as below: l110ae_rvt50_v021.mdl l110ae_rvt50_v021.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

5 V n-ch MOSFET
5 V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.2.4

For detailed descriptions of the compact model equations and parameters, please refer to this material.

BSIM3V3 MOSFET Model User Manual

In BSIM3V3.2.4 there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below, and the simulation results from other versions of the simulators might be different

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version 6.11_1.1

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (5 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo model
TT: Typical n-ch MOSFET and Typical p-ch MOSFET
SS: Total corner of Slow n-ch MOSFET and Slow p-ch MOSFET
SNFP: Total corner of Slow n-ch MOSFET and Fast p-ch MOSFET
FNFP: Total corner of Fast n-ch MOSFET and Slow p-ch MOSFET
FF: Total corner of Fast n-ch MOSFET and Fast p-ch MOSFET
TT_G: Typical n-ch MOSFET and Typical p-ch MOSFET (the same as TT)
SS_G: Global corner of Slow n-ch MOSFET and Slow p-ch MOSFET
SNFP_G: Global corner of Slow n-ch MOSFET and Fast p-ch MOSFET
FNFP_G: Global corner of Fast n-ch MOSFET and Slow p-ch MOSFET
FF_G: Global corner of Fast n-ch MOSFET and Fast p-ch MOSFET

1.3.3 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, the user can run Monte-Carlo mismatching simulation over a corner model. For example, the user can evaluate the mismatching performance at the SS_G case.

1.3.4 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.5 Model usage

The definitions of sub-circuit model parameters

After invoking Monte-Carlo model parameters, the compact transistor models evolve to sub-circuit models. The input parameters for the sub-circuit model are the same as the compact model, except two more input parameters (mis_flag, & mf) are added for the sub-circuit model.

The following is the parameter list:

- nf: number of gate fingers
- m: number of devices
- mf: the same as "m"
- mis_flag: the flag parameter is used to turn on/off mismatching parameters for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.
- w: total gate width (gate finger width * number of gate fingers)
- l: gate finger length
- as: source area per finger
- ad: drain area per finger
- ps: source periphery per finger
- pd: drain periphery per finger
- sa: distance between OD edge to poly from one side
- sb: distance between OD edge to poly from the other side
- sd: poly space for multiple finger device
- sca: integral of the first distribution function for scattered well dopant
- scb: integral of the second distribution function for scattered well dopant
- scc: integral of the third distribution function for scattered well dopant

The following example shows 3 transistors connected in parallel; each transistor has 5 fingers and each finger has width 5u and length 1u. The mismatching parameters are turned on. And the parameter "mf" is a must for mismatching simulation of parallel devices. Please note mf and m have to be equal.

```
x1 (d g s b) DeviceName w='W1*NF1' l=1u nf=5 mis_flag=1 m=3 mf=3
```

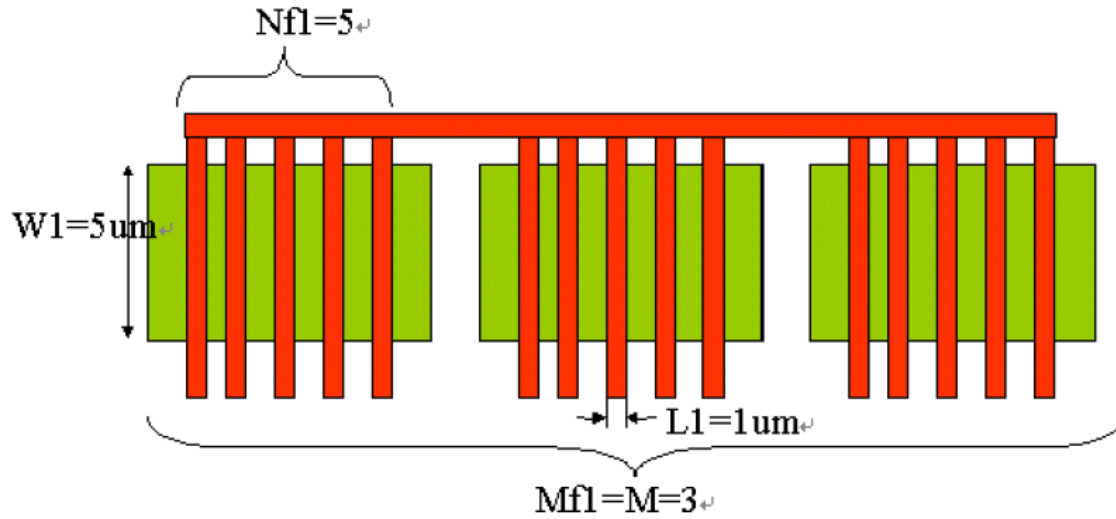
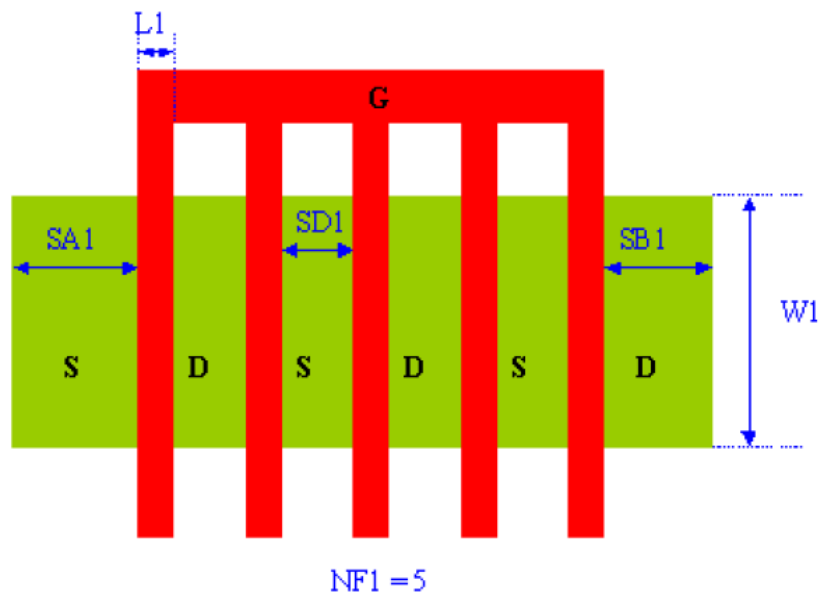
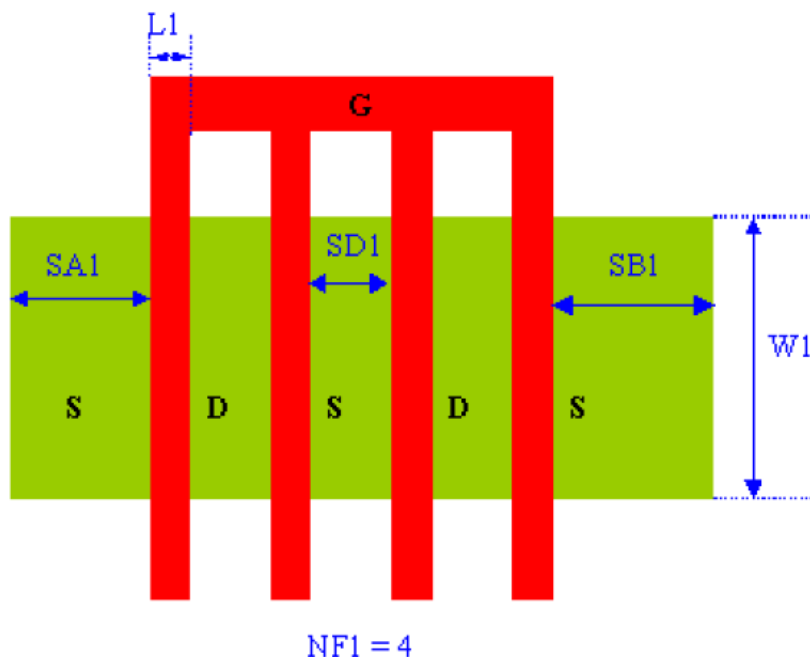



Figure 1-1 Layout Example of parallel devices.

Follow the description for the net-list of the LOD effect and these steps to invoke the model file during simulation.



(a) Example for odd finger number



(b) Example for even finger number

Figure 1-2 Layout Example for LOD effect

For Synopsys HSPICE

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

- Include library file and select section.

Example 1: .lib " l110ae_rvt50_v021_mc_corner.lib" MC

Example 2: .lib " l110ae_rvt50_v021_mc_corner.lib" FF

Example 3: .lib " l110ae_rvt50_v021_mc_corner.lib" FF_G

- Define sigma value for MC simulation.

Example: .param sigma=3

- Add the following statement in the input file to the simulator for corner cases

.lib "/ModelFileName.lib" CaseName

, where CaseName could be MC, TT, FF, SS, FNFP, SNFP, FF_G, SS_G, FNFP_G or SNFP_G.

- Define device condition by 'Xxx' statement.

Example: For Finger Number device (Refer to Figure 1-2)

```
X1 Vdd Vgg Vss Vbb DeviceName W='W1*Nf1' L=L1 NF=Nf1
+ SA=SA1 SB=SB1 SD=SD1 SCA=SCA1 SCB=SCB1 SCC=SCC1
+ NF_ODD='NF-2*INT(NF/2)' NF_EVEN='1-NF_ODD'
+ A_UNIT='SD*W/NF' P_UNIT='2*(SD+W/NF)'
+ A_SA_EDGE='SA*W/NF' P_SA_EDGE='2*(SA+W/NF)'
+ A_SB_EDGE='SB*W/NF' P_SB_EDGE='2*(SB+W/NF)'
+ AD='NF_ODD*(A_UNIT*((NF+1)/2-1)+A_SB_EDGE) +
NF_EVEN*(A_UNIT*Nf/2)'
+ AS='NF_ODD*(A_UNIT*((NF+1)/2-1)+A_SA_EDGE) +
NF_EVEN*(A_UNIT*(NF/2-1)+A_SA_EDGE+A_SB_EDGE)'
+ PD='NF_ODD*(P_UNIT*((NF+1)/2-1)+P_SB_EDGE) +
NF_EVEN*(P_UNIT*Nf/2)'
+ PS='NF_ODD*(P_UNIT*((NF+1)/2-1)+P_SA_EDGE) +
NF_EVEN*(P_UNIT*(NF/2-1)+P_SA_EDGE+P_SB_EDGE)'
```

- Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the global MC simulation

0 to turn off the global MC process simulation

MISMATCH: 1 to turn on the local MC mismatch simulation

0 to turn off the local MC mismatch simulation

Example 1:

run global & local (total) MC simulation together.

```
.lib "/ModelFileName.lib" MC
```

```
.PARAM PROCESS=1 MISMATCH=1
```

Example 2:

run global MC simulation only.

```
.lib "/ModelFileName.lib" MC
```

```
.PARAM PROCESS=1 MISMATCH=0
```

Example 3:

run local MC simulation only.

```
.lib "/ModelFileName.lib" MC
```

```
.PARAM PROCESS=0 MISMATCH=1
```

Example 4:

run local MC simulation on global corners.

```
.lib "/ModelFileName.lib" FF_G (or SS_G, FNFP_G, SNFP_G)
```

```
.PARAM PROCESS=0 MISMATCH=1
```

- Define the number of Monte Carlo iterations to run MC simulations.

Example: .DC Vgg 0 5 0.01 Sweep Monte=1000

For Cadence SPECTRE

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9

- Include library file and select section.

Example 1: include "l110ae_rvt50_v021_mc_corner.lib.scs" section=mc

Example 2: include "l110ae_rvt50_v021_mc_corner.lib.scs" section=ff

Example 3: include "l110ae_rvt50_v021_mc_corner.lib.scs" section=ff_g

- Define sigma value for MC simulation.

Example: parameters sigma=3

- Add the following statement in the input file to the simulator for corner cases.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be mc, tt, ff, ss, fnsp, snfp, ff_g, ss_g, fnsp_g or snfp_g.

- Define device condition by 'Xxx' statement.

Example: For Finger Number device (Refer to Figure 1-2)

```
X1 (Vdd Vgg Vss Vbb) DeviceName w='W1'*NF1' l=L1 nf=NF1
+ sa=SA1 sb=SB1 sd=SD1 sca=SCA1 scb=SCB1 scc=SCC1
+ nf_odd=nf-2*int(nf/2) nf_even=1- nf_odd
+ a_unit=sd*w/nf p_unit=2*(sd+w/nf)
+ a_sa_edge=sa*w/nf p_sa_edge =2*(sa+w/nf)
+ a_sb_edge=sb* w/nf p_sb_edge =2*(sb+w/nf)
+ ad=nf_odd *( a_unit *((nf+1)/2-1)+ a_sb_edge) + nf_even *( a_unit
*nf/2)
+ as=nf_odd *( a_unit *((nf+1)/2-1)+ a_sa_edge) + nf_even *( a_unit
*(nf/2-1)+ a_sa_edge + a_sb_edge)
+ pd=nf_odd *( p_unit *((nf+1)/2-1)+ p_sb_edge) + nf_even *( p_unit
*nf/2)
+ ps=nf_odd *( p_unit *((nf+1)/2-1)+ p_sa_edge) + nf_even *( p_unit
*(nf/2-1)+ p_sa_edge + p_sb_edge)
```

- Define the number of Monte Carlo iterations and control MC simulations.

Example 1: run global & local (total) MC simulation together.

mc1 montecarlo numruns=1000 seed=1 variations=all donominal=yes

savefamilyplots=yes {

analysisDC1 dc param=Vgg start=0 stop=5 step=0.01}

Example 2: run global MC simulation only

mc1 montecarlo numruns=1000 seed=1 variations=process donominal=yes

savefamilyplots=yes {

```
analysisDC1 dc param=Vgg start=0 stop=5 step=0.01}
```

Example 3: run local simulation only

```
mc1 montecarlo numruns=1000 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes {
```

```
analysisDC1 dc param=Vgg start=0 stop=5 step=0.01}
```

Example 4: run local MC simulation on global corners.

```
include "./ModelFileName.lib.scs" section= ff_g ( or ss_g, fnsp_g, snfp_g )
mc1 montecarlo numruns=1000 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes {
```

```
analysisDC1 dc param=Vgg start=0 stop=5 step=0.01}
```

For Mentor Graphics ELDO

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.Option scale=0.9
```

- Include library file and select section.

Example 1: .lib " l110ae_rvt50_v021_mc_corner.lib.eldo" MC

Example 2: .lib " l110ae_rvt50_v021_mc_corner.lib.eldo" FF

Example 3: .lib " l110ae_rvt50_v021_mc_corner.lib.eldo" FF_G

- Define sigma value for MC simulation.

Example: .param sigma=3

- Add the following statement in the input file to the simulator for corner cases.

```
.lib "./ModelFileName.lib.eldo" CaseName
```

, where CaseName could be MC, TT, FF, SS, FNFP, SNFP, FF_G, SS_G, FNFP_G or SNFP_G.

-Define device condition by 'Xxx' statement.

Example: For Finger Number device (Refer to Figure 1-2)

```
X1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1
+ SA=SA1 SB=SB1 SD=SD1 SCA=SCA1 SCB=SCB1 SCC=SCC1
+ NF_ODD='NF-2*INT(NF/2)' NF_EVEN='1-NF_ODD'
+ A_UNIT='SD*W/NF' P_UNIT='2*(SD+W/NF)'
A_SA_EDGE='SA*W/NF' P_SA_EDGE='2*(SA+W/NF)'
+ A_SB_EDGE='SB*W/NF' P_SB_EDGE='2*(SB+W/NF)'
+ AD='NF_ODD*(A_UNIT*((NF+1)/2-1)+A_SB_EDGE) +
NF_EVEN*(A_UNIT*NF/2)'
+ AS='NF_ODD*(A_UNIT*((NF+1)/2-1)+A_SA_EDGE) +
NF_EVEN*(A_UNIT*(NF/2-1)+A_SA_EDGE+A_SB_EDGE)'
+ PD='NF_ODD*(P_UNIT*((NF+1)/2-1)+P_SB_EDGE) +
NF_EVEN*(P_UNIT*NF/2)'
```

+ PS='NF_ODD*(P_UNIT*((NF+1)/2-1)+P_SA_EDGE) +
NF_EVEN*(P_UNIT*(NF/2-1)+P_SA_EDGE+P_SB_EDGE)'

- Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the global MC simulation
0 to turn off the global MC process simulation
MISMATCH: 1 to turn on the local MC simulation
0 to turn off the local MC simulation

Example 1:

run global & local (total) MC simulation together.

.lib "/ModelFileName.lib.eldo" MC
.PARAM PROCESS=1 MISMATCH=1

Example 2:

run global MC simulation only.

.lib "/ModelFileName.lib.eldo " MC
.PARAM PROCESS=1 MISMATCH=0

Example 3:

run local MC simulation only.

.lib "/ModelFileName.lib.eldo " MC
.PARAM PROCESS=0 MISMATCH=1

Example 4:

run local MC simulation on global corners.

.lib "/ModelFileName.lib.eldo " FF_G (or SS_G, FNFP_G, SNFP_G)

.PARAM PROCESS=0 MISMATCH=1

- Define the number of Monte Carlo iterations to run MC simulations.

Example: .DC Vgg 0 5 0.01 Sweep Monte=1000

Benchmark Model: l110ae_rvt50_v021

DeviceName for 5 V n-ch MOSFET: n_50_rvtl110ae

DeviceName for 5 V p-ch MOSFET: p_50_rvtl110ae

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range (After 90% shrinkage)

For both 5 V n-ch and p-ch MOSFET:

0.488 um <= L_{DES} <= 10 um
0.350 um <= W_{DES} <= 10 um
0.324 um <= SA, SB

Note: It is recommended to adopt multi-finger structures if $W_{DES} > 10 \mu m$.

Adopt multi-finger structures if $W_{DES} > 10 \mu m$.

$0.324 \mu m \leq SA, SB \leq 2 \mu m$

Note: $SAREF = SBREF = 3.6 \mu m$ is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 5 V n-ch MOSFET:

$0V \leq V_{GS} \leq 5V (*1.1)$

$0V \leq V_{DS} \leq 5V (*1.1)$

$-3V \leq V_{BS} \leq 0V$

For 5 V p-ch MOSFET:

$0V \geq V_{GS} \geq -5V (*1.1)$

$0V \geq V_{DS} \geq -5V (*1.1)$

$3V \geq V_{BS} \geq 0V$

Temperature range:

$-40C \sim 125C$

Feature\ GDSII	Drawing in shrinkable L130AE GDS database (unit : um)	L110AE GDSII =0.9*(L130AE GDS) Dimension (unit : um)	L110AE SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	HS(L + 8 nm): 9/0.116 LL/SP(L + 12 nm): 9/0.12
2.5V I/O (W/L)	10/0.24	9/0.216	(L + 12 nm): 9/0.228
3.3V I/O (W/L)	10/0.34	9/0.306	(L + 12 nm): 9/0.318
5V I/O (W/L)	10/0.542	9/0.488	(L + 12 nm): 9/0.5
HG 3.3V I/O (W/L)	HG IO-PMOS : 10/0.32 HG IO-NMOS : 10/0.34	HG IO-PMOS : 9/0.228 HG IO-NMOS : 9/0.306	HG IO-PMOS (L + 12 nm): 9/0.3 HG IO-NMOS (L + 12 nm): 9/0.318
Salicide and Non-salicide resistor	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Mixed-Mode Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
RF Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Contact size	0.16*0.16	0.144*0.144	0.144*0.144
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	MVA1(W -0.02um) : 0.16*0.2 Other 1X MVA : 0.18*0.18 2X MVA (W -0.04um): 0.28*0.44
Metal size	Width/Space	0.9*(Width/Space)	0.9*(Width/Space)
Slotting inserting	Metal line width>5.556um	Metal line width>5um	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 Monte Carlo DC Simulation Results for Process Variation

One thousand Monte Carlo iterations are performed in the following simulation conditions, and the following corner points show TT, SS, FF, SNFP and FNSP values individually.

A reasonable number of Monte Carlo iterations is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to $(\text{iteration times})^{(-1/2)}$.

2.1 Threshold voltage in linear region VTLIN

The following figures show the Monte Carlo simulation results of linear threshold voltage extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS at 25 C.

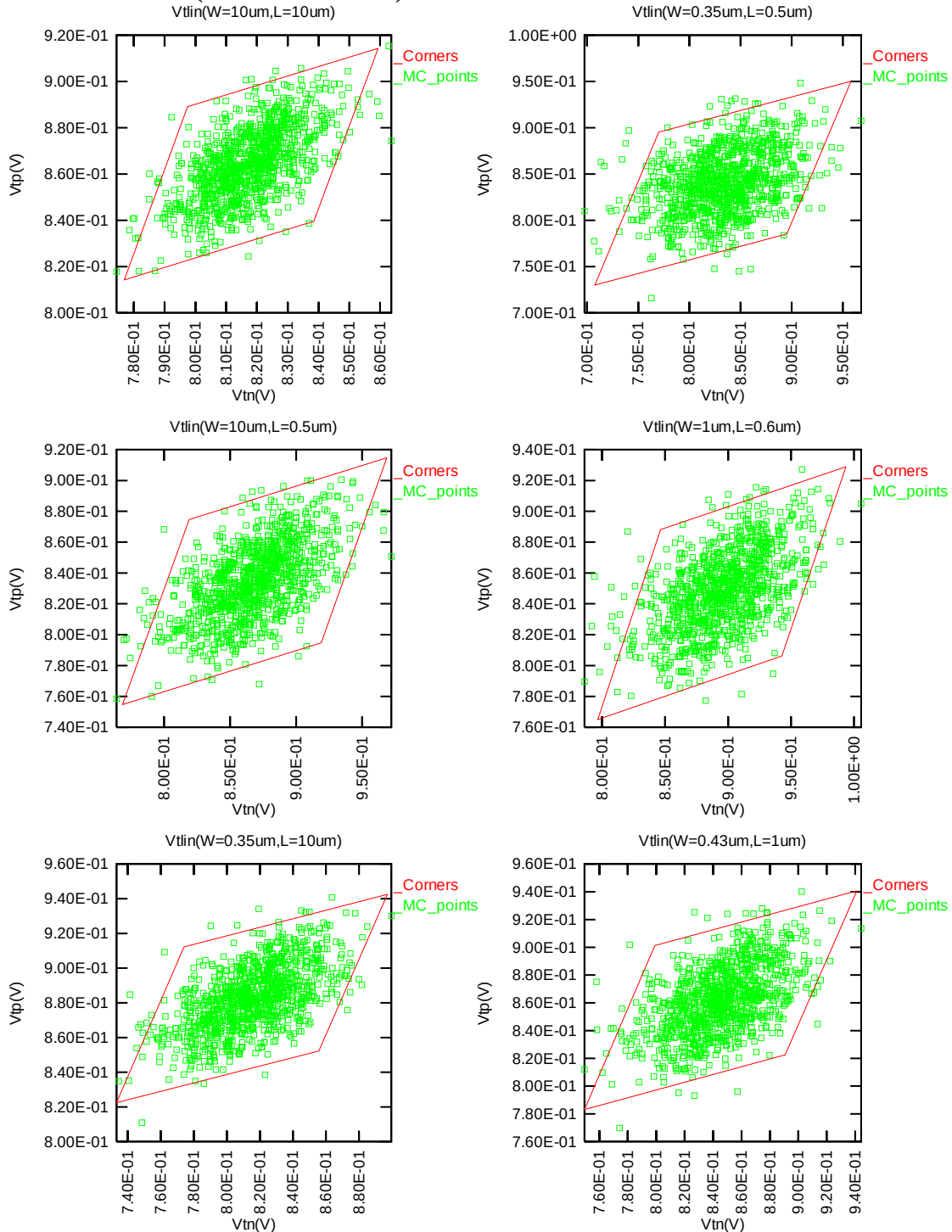


Figure 2-1 Monte Carlo plots of threshold voltage at low drain bias

2.2 Threshold voltage in saturation region VTSAT

The following figures show the Monte Carlo simulation results of saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS at 25C.

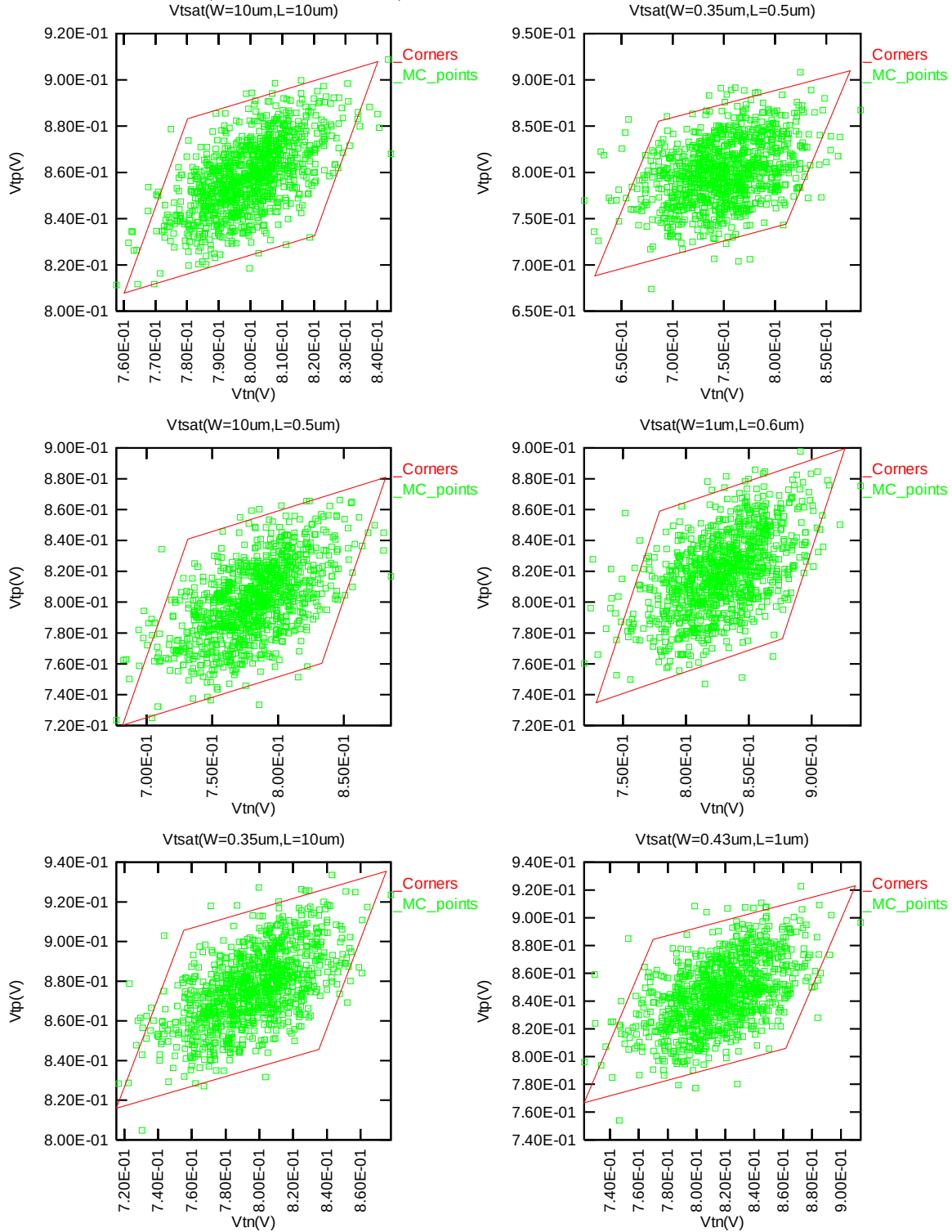


Figure 2-2 Monte Carlo plots of threshold voltage at high drain bias

2.3 Linear Current IDLIN

The following figures show the Monte Carlo simulation results of linear current at 25C.

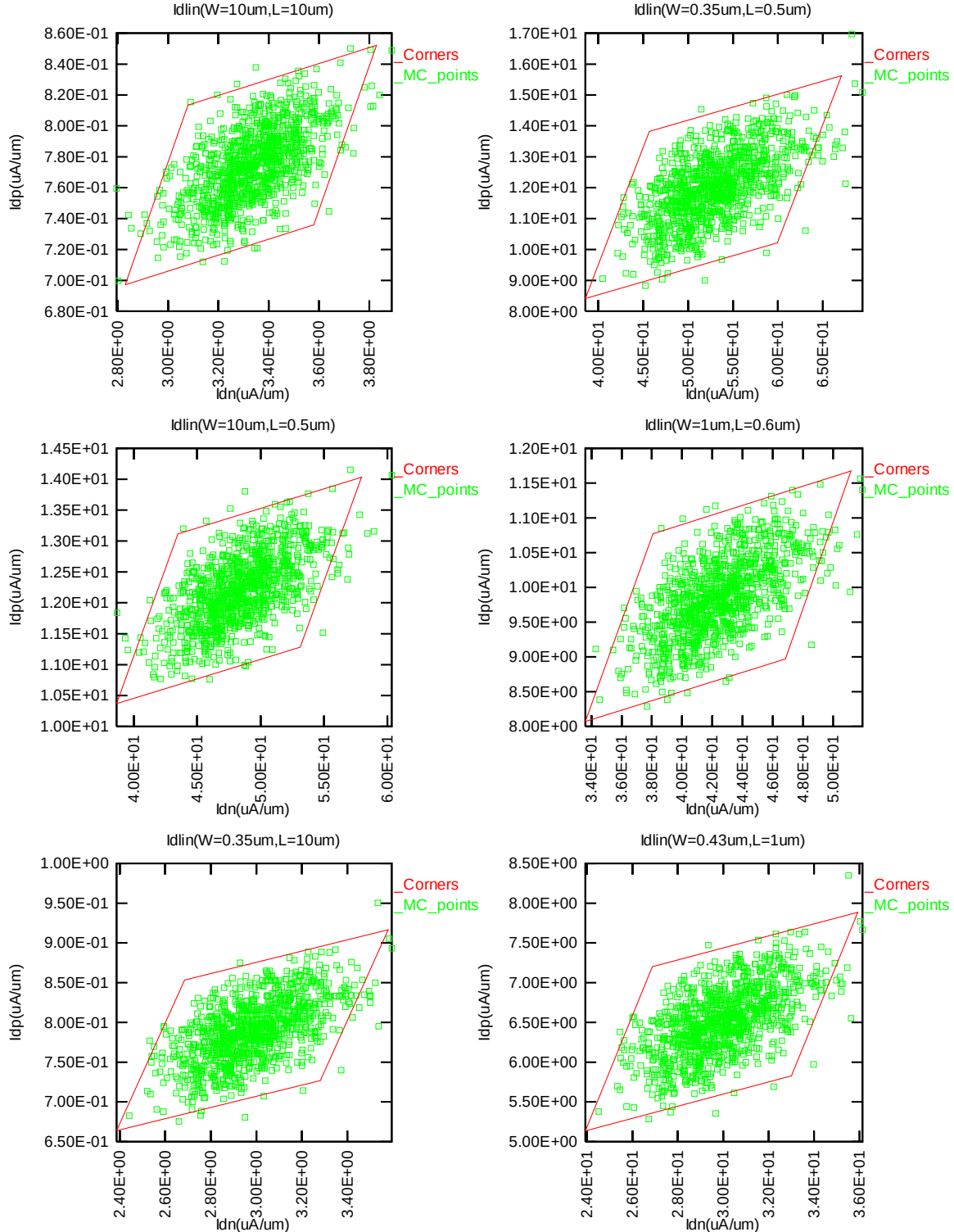


Figure 2-3 Monte Carlo plots of linear current

2.4 Saturation Current IDSAT

The following figures show the Monte Carlo simulation result of saturation current at 25C

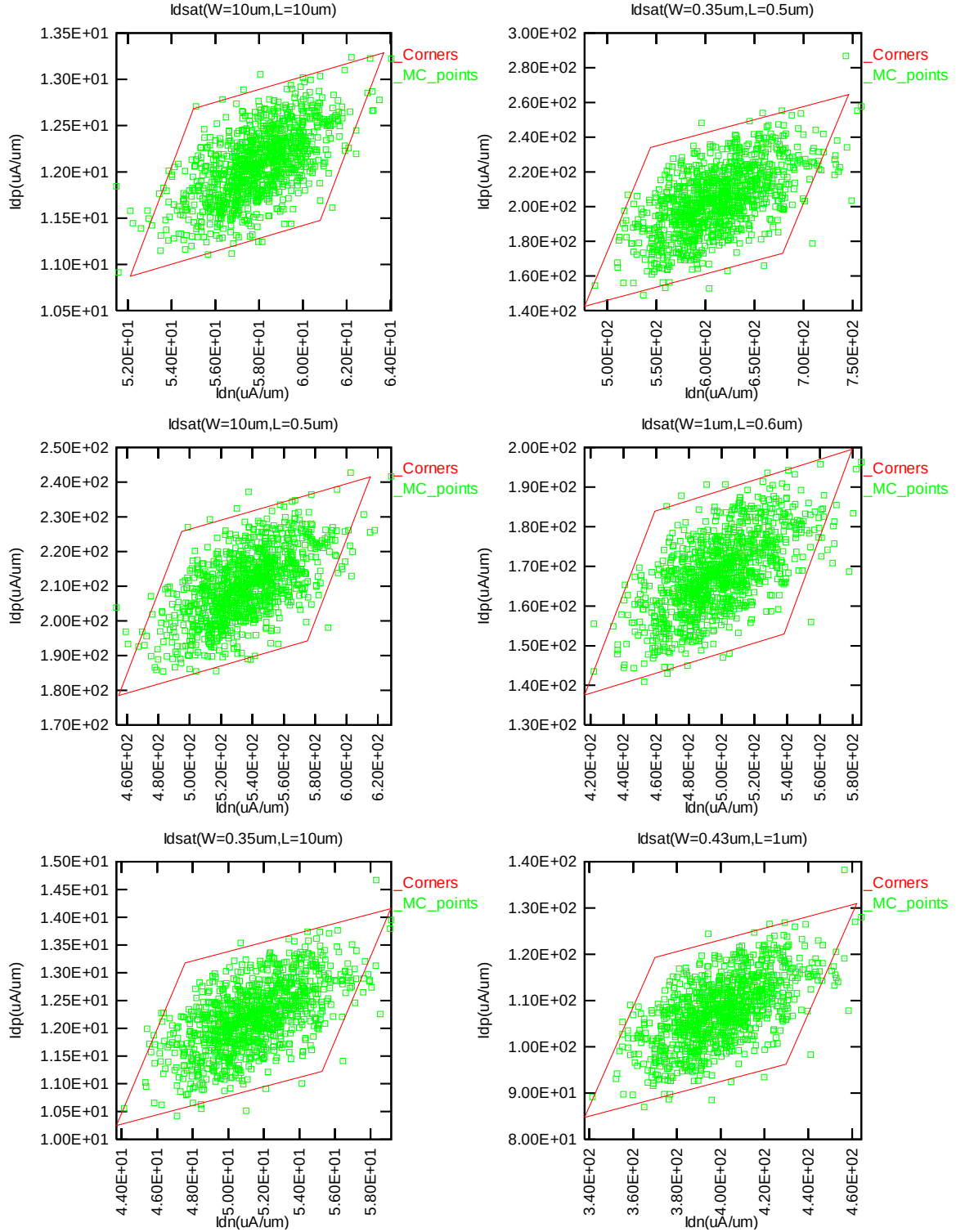


Figure 2-4 Monte Carlo plots of saturation current

2.5 Transfer Conductance GM

The following figures show the Monte Carlo simulation results of transfer conductance at 25C.

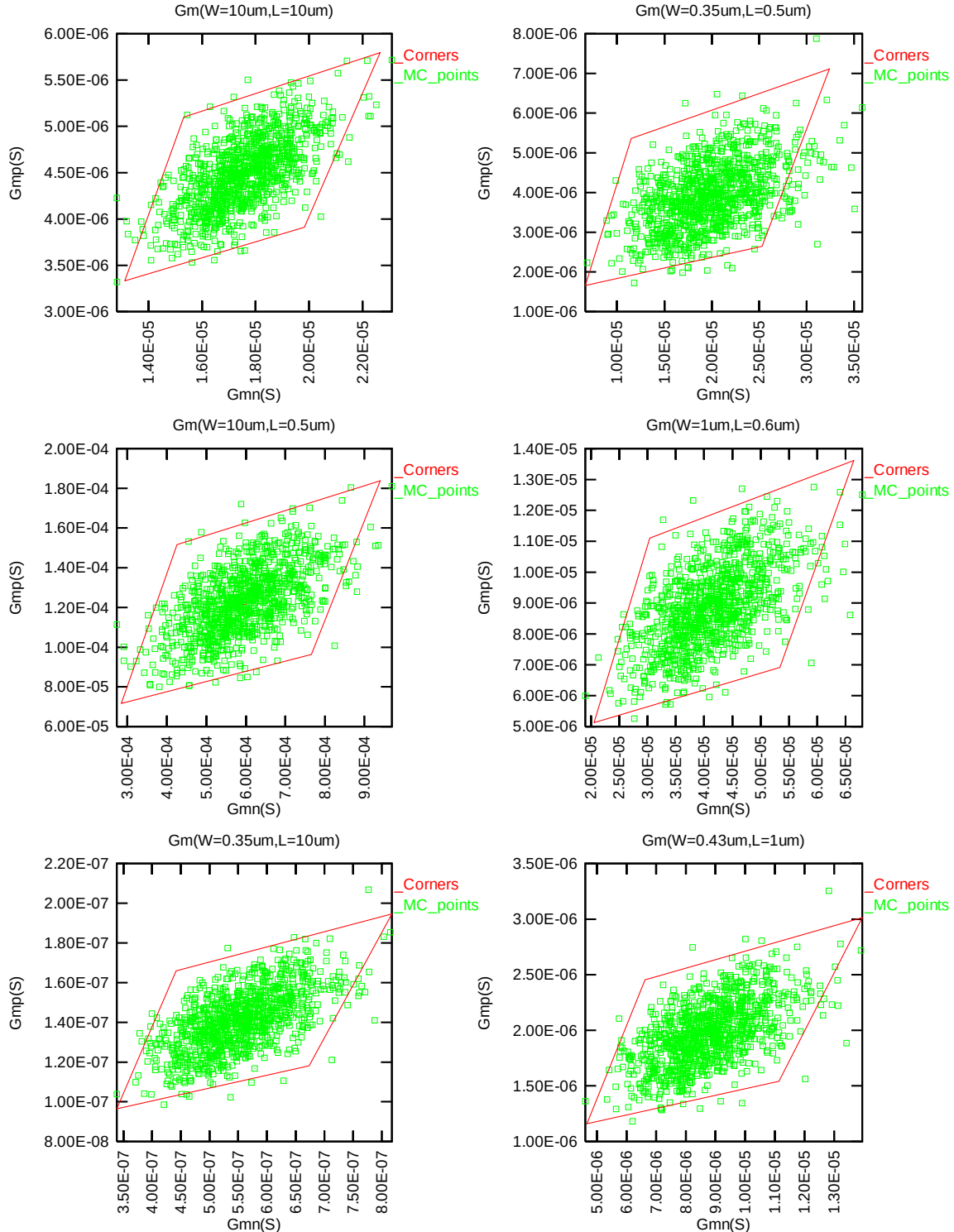


Figure 2-5 Monte Carlo plots of transfer conductance

2.6 Output Conductance GDS

The following figures show the Monte Carlo simulation results of output conductance at 25C.

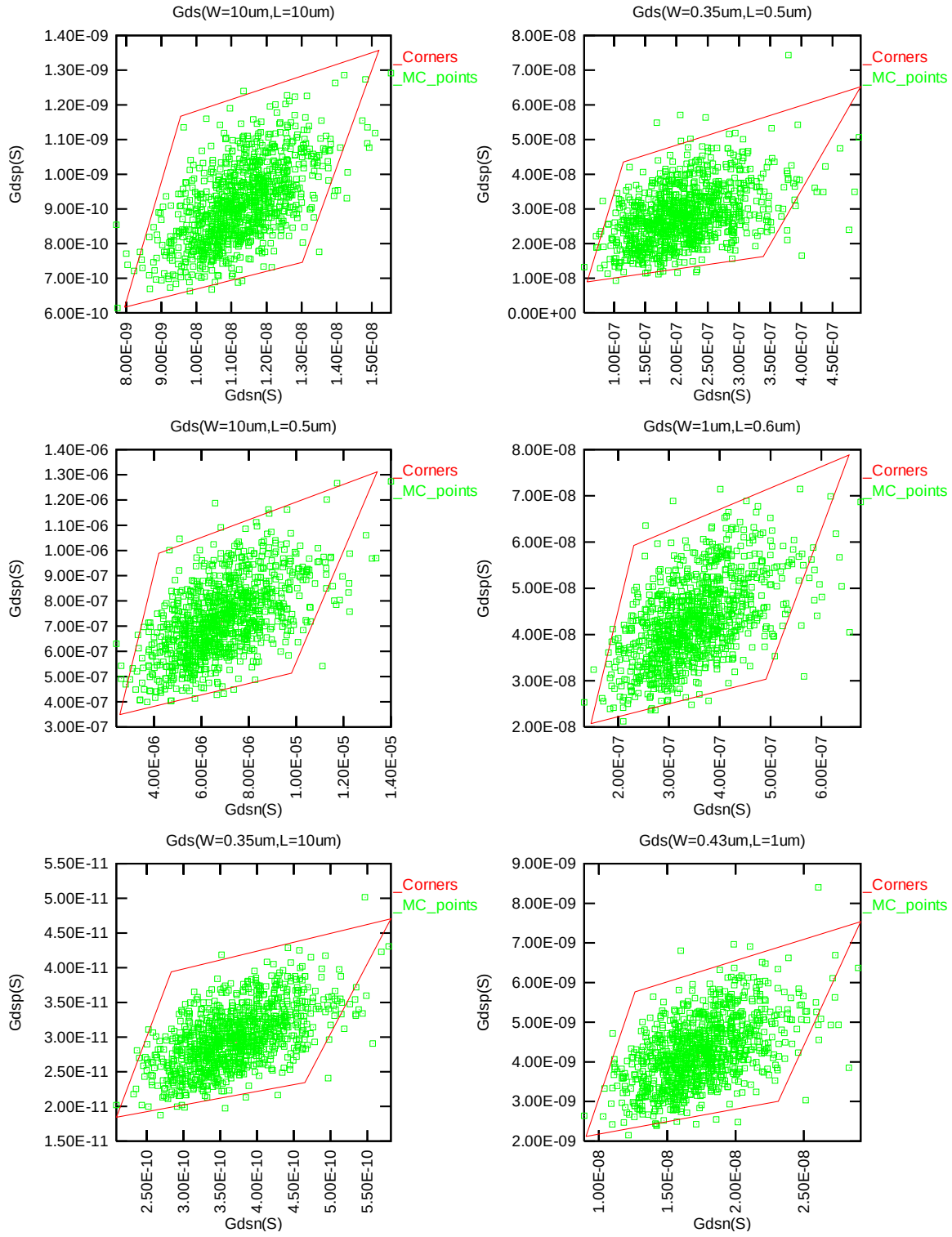


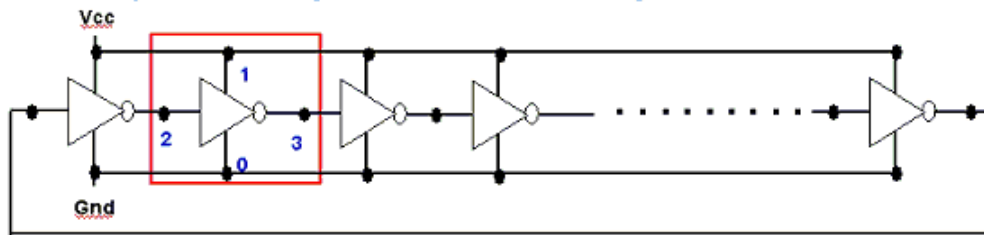
Figure 2-6 Monte Carlo plots of output conductance

3 Monte Carlo Dynamic Simulation Results for Process Variation

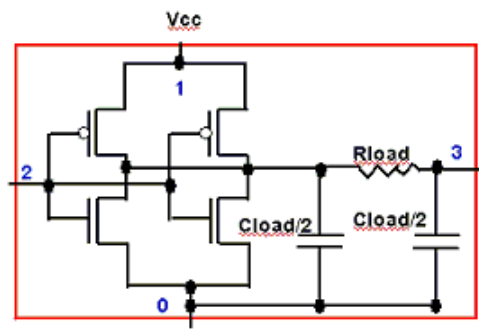
3.1 Ring Oscillator Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configuration is adopted

R.O. (Inverter) circuit description



Ring Oscillator circuit



single stage R.O. circuit

Figure 3-1 Ring oscillator netlist diagram

Table 3-1 Ring oscillator (inverter) structure description.

R.O. (Inverter) Structure description		
Contact size	0.144 um X 0.144 um	
Polysilicon gate to contact spacing	0.14 um	
Contact to diffusion edge spacing	0.06 um	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	2.5 um X 2	
Channel length for n-ch MOSFET	0.5 um	
Channel width for p-ch MOSFET	3.75 um X 2	
Channel Length for p-ch MOSFET	0.5 um	

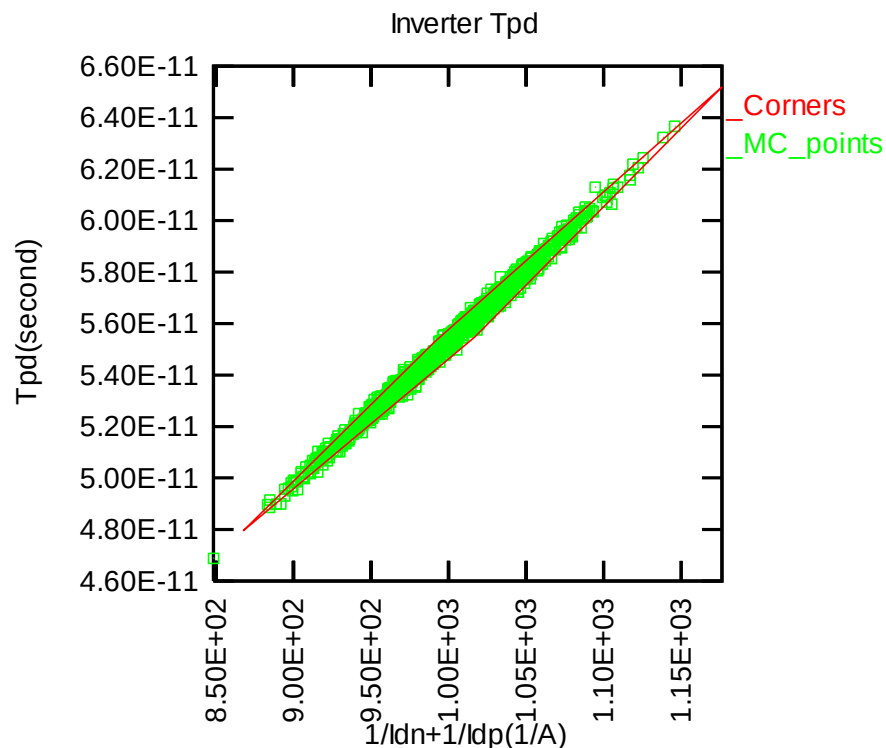


Figure 3-2 Corner plot of ring oscillator

4 Monte Carlo Simulation Results for Mismatch

4.1 Threshold voltage in linear region VTLIN

The following figures show the Monte Carlo mis-matching simulation results of linear threshold voltage extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS at 25C.

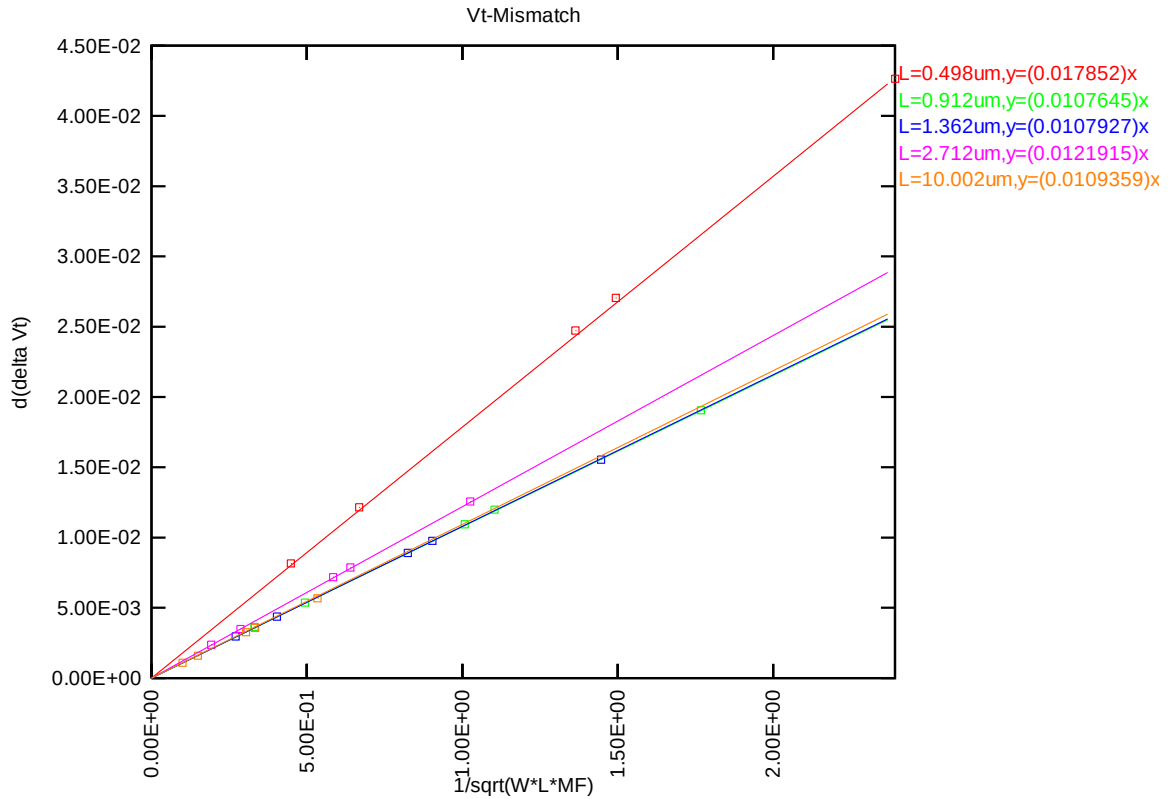


Figure 4-1 VTLIN mis-matching simulation of 5 V n-ch MOSFET

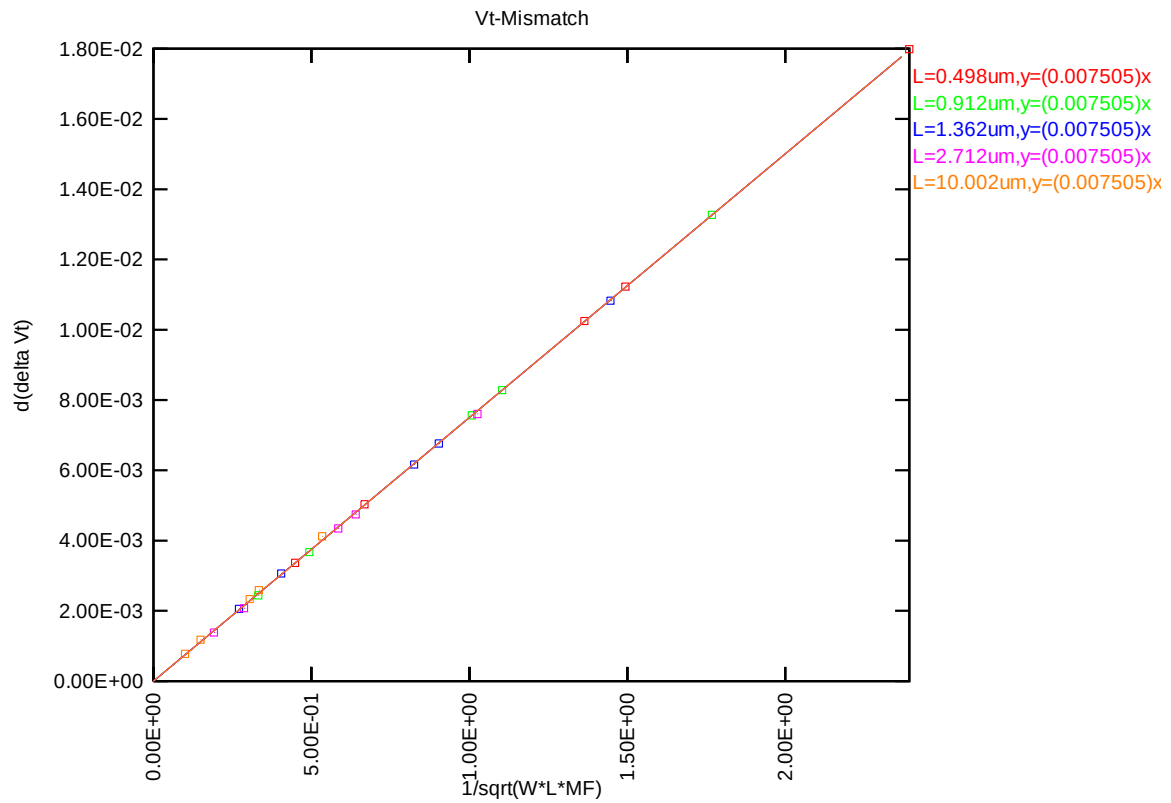


Figure 4-2 VTLIN mis-matching simulation of 5 V p-ch MOSFET

4.2 Saturation Current IDSAT

The following figures show the Monte Carlo mis-matching simulation results of saturation current at 25C.

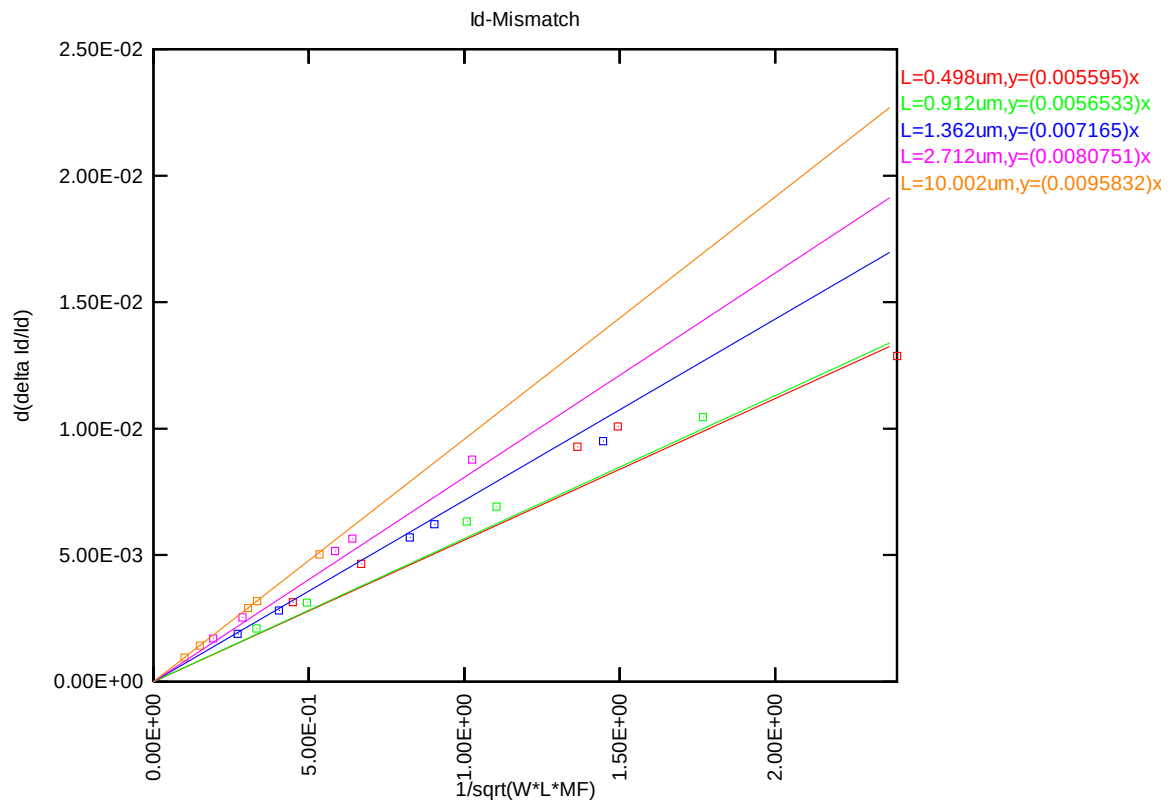


Figure 4-3 IDSAT mis-matching simulation of 5 V n-ch MOSFET

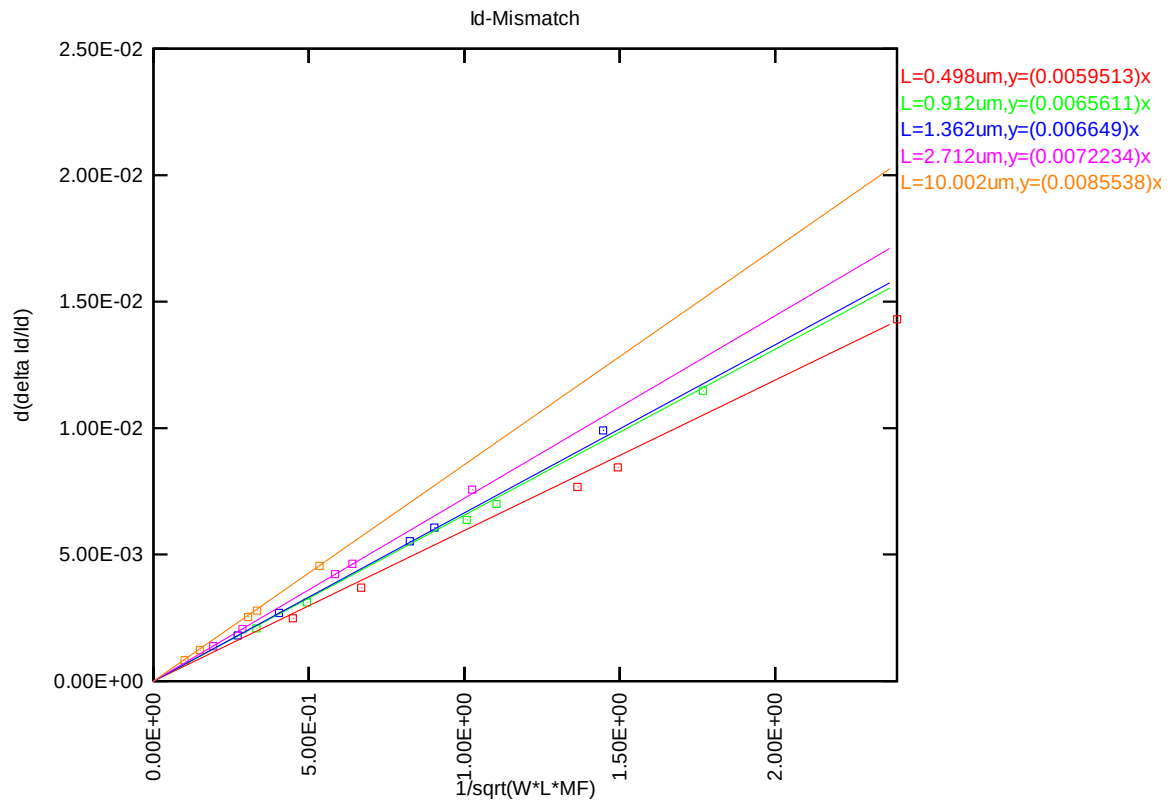


Figure 4-4 IDSAT mis-matching simulation of 5 V p-ch MOSFET

5 Appendix

5.1 HSPICE Warning Messages

Table 5-1 HSPICE warning messages

NO.	Warning Message	Explanation
1.	-	-

5.2 ELDO Warning Messages

Table 5-2 Eldo warning messages

NO.	Warning Message	Explanation
1.	-	-

5.3 SPECTRE Warning Messages

Table 5-3 Spectre warning messages

NO.	Warning Message	Explanation
1.	-	-